CURRICULUM VITAE

OF

ENGR. DR. AWUA JUSTIN TARKA

EDUCATIONAL QUALIFICATION

Doctor of Philosophy (Ph.D); Mechanical Engineering

Joseph Sarwuan Tarka University, Makurdi, Nigeria - **2023**

Master of Engineering (M.Eng) Thermofluid and Energy Studies

Federal University of Agriculture Makurdi – Nigeria - **2016**

Bachelor of Engineering (B.Eng); Mechanical Engineering (Second Class Honours, Upper Division), Federal University of Agriculture, Makurdi – Nigeria - **2003**

Senior School Leaving Certificate

Mount Saint Gabriel's Secondary School, Makurd**i** - **1995**

First School Leaving Certificate

Army Children School, Bukavu Barracks, Kano - **1988**

CURRENT RESEARCH INTEREST/EXTENSION

Application of ANSYS fluent Computational Fluid Dynamics in nanofluid modelling for heat transfer enhancement in cavity flow settings which is directly applicable in enhancement of cooling in automobiles and production industries.

Valuation and Evaluation of plants and Machineries of industries

PROFESSIONAL QUALIFICATIONS

Registered Engineer: Council for the Regulation of Engineers in Nigeria **(COREN) R.25,129**

Member; Nigerian Institution of Mechanical Engineers **(MNiMeCHE)**

Member: Nigerian Society of Engineers **(MNSE)**

TRAININGS ATTENDED/ CERTIFICATES ACQUIRED

- CERTIFICATE IN BASIC HUMAN RESOURCES MANAGEMENT PROGRAMME
- CERTIFICATE IN BASIC PUBLIC RELATIONS PROGRAMME
- CERTICATE IN GRANT WRITING PROPOSAL WORKSHOP (CIPESS)
- CERTIFICATE OF BEST CONFERENCE PAPER (POSTER) PRESENTATION, INTERNATIONAL CONFERENCE ON NANOTECHNOLOGY, UNIVERSITY OF NIGERIA NSUKKA.

PERSONAL PROFILE

Date of Birth:	-	28th August 1978
Marital Status:	-	Married
State of Origin	-	Benue State
Local Government Area	-	Gwer West
Sex:	-	Male

WORK EXPERIENCE

University

1st October 2023 - **Senior Lecturer (Thermofluid and Energy Studies)**
Mechanical Engineering Department
Joseph Sarwuan Tarka University, Makurdi

1st October, 2019 - **Lecturer I (Thermofluid and Energy Studies)**
Mechanical Engineering Department
Joseph Sarwuan Tarka University, Makurdi

17th November, 2016 - **Lecturer II**
Mechanical Engineering Department
Federal University of Agriculture, Makurdi.

22nd May 2012 – 17th November, 2016 - **Graduate Assistant**
Mechanical Engineering Department
Federal University of Agriculture Makurdi

Industry

Bezaleel+Turnkey Consultants Inc. Monrovia **2022 – Till Date**
Position: Plant and Machinery Consultant
Duties: Evaluation of Plant and Machinery of Firestone Company

Target Consult, Makurdi **2012 - Till Date**
Position: Consultant
Duties: - Checking and sealing of Mechanical Engineering Drawings
Supervision of Building Services works in the Field

Knight Frank, Lagos

Position Held: **Plant and Machinery Valuer/Analyst** **2010 - 2012**

Duties: -Inspection and valuation of client's plant and machinery of all kinds (which includes high capacity Electric power generating plant, factory production plant, services plant and motor vehicles)
-Physical and technical verification of plant and Machinery based on maintenance culture
-Analysis of maintenance schedule, operation procedures and purchase documents of machineries
-Inventory collection of plant and machinery, motor vehicles office equipment, furniture and fittings
-Inspection of all corrective and preventive maintenance records of plant and machinery and shipping documents.
-Analysis and quality verification of the working principles of all types of plant and machineries.
-Valuation of all types of plant and machinery based on open market value and forced sale values,
-Preparation of inventory schedule for all types of plant and machinery.
-Preparation of report on the basis of valuation using invoices, shipping documents and custom duties
-Sourcing for current market value of plant and machinery, office equipment, furniture and fittings.
-Documentation of valuation reports and other related materials.
-Gathering of information on current market trend and insurance policies in valuation for insurance Purpose.
- Supervision of the general maintenance of motor vehicles and other fleet systems.
-Maintenance of building and factory services plants and machineries.
-Logistics management and general operations of company fleet.

Achievements/Accomplishments: Inspected and valued the plant and machinery assets of the following companies:

- Nigeria Bottling Company (**NBC**) Lagos and Port-Harcourt
- **M.R.S Oil and Gas Jetty in Apapa Tincan Island Lagos**
- BUA Flour Mills Apapa Lagos
- Globe Spinning Industry Lagos
- ORC Fish Processing Company **(CHIVITA GROUP),** Lagos.
- Vitamalt PLC Agbara Estate Lagos.
- Union Forest Calabar, Cross River State
- BUA Sugar Refinery Apapa Lagos
- Edo Cement Okpella Edo State
- Unique Pharmaceuticals, Otta, Ogun State
- Ugor- Bas Construction Company Lagos
- Netcom Africa Telecommunication Lagos
- **Seplat Petroleum Development Company (Flowstations and Gas Plants) Delta State.**
- Allied Atlantic Distilleries Sango Otta, Ogun State
- MRS filling stations nation wide

ZEBRA TECHNICAL SERVICES LIMITED **2008 - 2010**
Position held: Regulatory Compliance/Project Service Engineer
(with Shell Nigeria Exploration and Production Company)

Duties: -management of **Bonga North West** Project compliance register/database
-Timely commencement of permits and consents acquisition process
- Following up with regulators to ensure timely approvals of projects permits and consents
-Liaison with regulatory agencies unbehalf of **SNEPCo Bonga North West** project
-Providing weekly status report to the Bonga North West Project team.
-Maintaining secure custody for all received permits and consents documents
- Supporting **Bonga North West** as ECMT administrator/superuser

-Updating the **SNEPCO Bonga North West** ECMT with new laws/guidance so that compliance strategy is based on current requirement.
-Assisting the project team in demonstrating compliance due diligence to regulators
-Conducting Training for Bonga Northwest project team members on the use of the ECMT
-Liaising with external regulatory agencies on permits and consents matters
-Assisting in the Development of project master schedule for Bonga northwest
-Assisting in the analysis of project Schedule and planning
-Assisting in project planning and analysis of contractual documents
-Assisting in the development of new project ideas and time saving strategies in liaison external regulators

Achievements/Accomplishments

1. Closed out the following milestone approvals on **OML-118 & OPL-212** (**Bonga Northwest project**)
-Approval for Field Development Plan from **DPR**
-Approval for Conceptual design
-Approval for Initial, Draft and Final Environmental Impact Assessment from **DPR** and **FMEnv**.
-Permit To Survey pipeline route from **DPR** and **NPA**

2. Developed a compliance strategy for detailed design of Bonga Northwest project surface facility
3. Developed an engagement strategy for strengthening regulatory compliance stakeholder relations
4. Conducted training on a web based regulatory compliance software for Bonga Northwest project team members
5. Facilitated the adequate representation of external regulators in all Subsea, Topsides and Umbilical Hazops
6. Prepared a compliance matrix capturing all regulatory compliance project activities on the project
7. Cross linked all regulatory compliance project items with their individual applicable regulations.
8. Facilitated the design, construction and management of the **DPR** Compliance Monitoring system
9. Facilitated the design and development of the **NNRA** oil and gas regulatory compliance system

RELIANCE GLOBAL TRENDS INVESTMENTS LTD **2007**
Position held – Technical Manager
Duties: specification Design, Editing and Certification of graphics (using coral draw package)
Writing, submitting and pursuant of proposal for approval
General maintenance of Universal Laser engraving machine
Supervision of workers in the technical department
Assembling and customization of materials to recommended

FAB TECH. NIG. LTD. **2006**
Position held – Technical Supervisor
Duties: Supervision of supply, installation, certification and commissioning of generators
Fabrication of industrial shear butter buckets and decanters.
Supervision of workers in technical department

ST. JOSEPH'S SCIENCE AND TECHNICAL COLLEGE MARKUDI **2005**
Position held – SS2&SS3 Physics Tutor
Duties: Taught full syllable in Physics

FEDERAL UNIVERSITY OF TECHNOLOGY, YOLA (NYSC) **2004**
Position Held - Graduate Assistant Lecturer
Duties: Taught applied mechanics and Engineering Drawing

NIGERIA BOTTLING COMPANY (NBC) MARKUDI **2001**
Position held – Technical Maintenance Assistant(SIWES)

Duties: Supervision of workers on production line
General maintenance of the washer and the filler machines
Generator Maintenance

NIGERIAN NATIONAL PETROLUEM CORPORATION (NNPC) ABUJA. (SIWES) **2001**
Position Held – Technical Maintenance Assistant
Duties: Maintenance of central air conditioning system

UNIVERSITY TEACHING EXPERIENCE

Energy Conversion (EME 515)

Numerical Analysis (EME 527)

Machine Shop Technology (EME 306)

Fluid Mechanics I (EME 202)

Fluid Mechanics II (EME 303)

Engineering Thermodynamics I (EME 205)

Introduction to Complex Analysis (EMA 201)

Transforms and probability (EMA 312)

Fluid Mechanics I (GET 205)

UNIVERSITY ADMINISTRATIVE EXPERIENCE

Post Graduate Students Coordinator - 19th May, 2023 - Till Date

SPECIAL SERVICE TO THE UNIVERSITY

College of Engineering Representative

To College of Vetenary Medicine

Joseph Sarwuan Tarka University, Makurdi - 2024

NATIONAL SERVICE

Ward Collation Officer: Logo and Gwer West Local Government; 2015 general election

Ward Collation Officer: Kwande House of Representative elections in 2019 elections

STUDENTS SUPERVISED

PGD

1. Ali Adamu (18/11587/PGD). Performance Evaluation of Mixed Mode Solar Dryer with Black Painted Pebbles as a Storage Material Using Energy Analysis
2. Entonu Awodi (19/12963). Evaluation of Thermal Properties of Some Common Roofing Materials in Nigeria.
3. Onuh Lawrence (19/12959/PGD. Conversion of automobile Engine Transmission System into a Training Model for Automobile Technology
4. Seki-JohnBull Eferero Biokpo (17/10626/PGD). Assesment of Greenhouse Gas Emission of Treasure Hotel/Restaurant in Portharcourt, Nigeria.
5. Okan Williams (15/8020/M.Eng) Performance Evaluation of a Solar Domestic Hot Water Thermosyphon System for Gembu Taraba State Nigeria.

B.Eng

1. Akatu Sylvester Adakole (16/37266/UE) and Ayom Cletus Aondohemba (16/37286/UE). Investigation of Maintenance Culture of Mechanical Workshop and Laboratory for Effective Instructional Delivery; Case Study of Joseph Sarwuan Tarka University, Makurdi.
2. Igbinidu Osaze Moses (16/37307/UE). Energy Audit of Selected Commercial Banks in Makurdi, Benue Sate.
3. Adio Stephen Adebayo (15/32864/UE) and Ojobo Samuel Inalegwu (15/32926). Evaluation of the Energy Consumption Pattern of Commuter Buses in Makurdi.
4. Enejoh, Moses Odufu (15/32894/UE) and Ochonu, Peter Enenche (15/32919/UE). Determination of the Heat Transfer Enhancement Potentials of Selected Green Nanofluid Produced from Flamboyant Tree Bark Nanoparticles.
5. Hiikya Benjamin Terhemba (14/28957/UE) and Offikwu Benedict Sule (14/28951/UE). Effect of Stale Cassava on Biogas Production Using Pig Waste.
6. Apa Jonathan Chijioke (18/46164/UE) and Amana Ojonugwa Daniel (19/52236/DE). Production and Evaluation of Biogas Using Vegetable/Watermelon Waste and Stagnated Drainage Water.
7. Okoh Benjamin Toochukwu (18/46124/UE) and Okolo Joshua Odewu (18/46203/UE). Evaluation of Physical Properties of Hybrid Bio-Nanofluid Produced from Flamboyant and Afara Tree Barks.

PUBLICATIONS WITH DATES

Dessertations and Theses

Justin T. Awua and Emmanuel Oko. (2003). Effect of Partially Substituting Cattle Dung with Piggery Manure on Biogas Generation. Bachelor of Science Degree project. Department of Mechanical Engineering, Federal University of Agriculture, Makurdi.

Awua Justin T. (2016). Determination of Thermophysical Properties of Nanofluid Prepared from Palm Kernel Fibre Nanoparticles. M.Eng Thesis. Federal University of Agriculture, Makurdi.

Awua Justin T. (2023). Evaluation of the Heat Transfer Properties of Nanofluid Produced from Afara and Flamboyant tree bark. Ph.D Thesis. Joseph Sarwuan Tarka University, Makurdi.

JOURNAL PAPERS

1. **Justin T, Awua,** Jacob S, Ibrahim. Saheed A Adio, Mehdi Mehrabi, Mohsen Sharifpur and Josua P Meyer (2018). Experimental investigations into viscosity, pH and electrical, conductivity of nanofluid prepared from palm kernel fibre and a mixture of water and ethylene glycol. *Bull. Mater. Sci. Indian Academy of Sciences.41:156.* *https://doi.org/10.1007/s12034-018-1676-1*.

2. **Justin Awua,** Sunday Ibrahim and Aondona Kwaghger, (2017). Investigation of Thermal Conductivity of Palm Kernel Fibre Nanofluid Using De-Ionized Water and Ethylene Glycol. Mixed at Ratio of 50:50 and 60:40. *International Journal of Engineering Trends and Technology (IJETT*) – Volume 49 No 1. DOI. 10.14445/22315381/IJETT-V49P207. http://www.ijettjournal.org

3. Alex Okibe Edeoja and **Justin Awua Tarka, (2018).** Effect of Penstock Diameter of a Simple Pico Hydro System on Shaft Power. *Elixir International Journal.* Elixir Mech. Engg. 118 50611, 50610-50622. www.elixirpublishers.com.

4. Edeoja, Alex Okibe and **Awua, Justin. (2018).** Shaft Power of a Simple Pico Hydropower System as A Function of The Flow Rate and Net Head Product. *International Journal of Engineering Technology Research & Management, Vol (02) _Issue (03). (http://ijetrm.com/).*

5. **J. T. Awua**, J. S. Ibrahim, M. Sharifpur and J. P. Meyer. **(2022).** Particle Characterization and Stability of Nanofluid Prepared from Palm Kernel Fibre with Mixture of Water and Ethylene Glycol as Base Fluid. *Journal of Mechanics Engineering and Automation 12 23-28 doi: 10.17265/2159-5275/2022.01.003.*

6. **Awua, Justin Tarka**, Sunday Ibrahim, Adama Lukeman and Gbugh Kenneth. (2023). Assessment of Energy Consumption Pattern of Local Palm Oil Production in ITU and Abak Local Government Area of Akwa Ibom State. Nigeria. *International Journal of Advances in Engineering and Management (IJAEM) Volume 5, Issue, pp: 474-480 www.ijaem.net ISSN:* 2395-5252. www.ijaem.net

7. **J. T. Awua**, J. S. Ibrahim, A. O. Edeoja, A. Kuhe, K. T Gbugho, M. Sharifpur. **(2023).** Influence of Volume Concentration and Temperature on Viscosity and Thermal Conductivity of Terminalia Superba (Afara) Tree Bark Nanofluid. *International Journal of Science and Research (IJSR). Volume 12 Issue 6, June 2023 ISSN: 2319-7064 SJIF 7.942.* **www.ijsr.net.**

8. **Awua J.T**., Kabiru B. and L. A Enyejo. (2023). Evaluation of Physical and Combustion Characteristics of Briquettes of Palm Fruit Fibre and Maize Chaff Using Cassava Extract as Binder. *World Journal of Advanced Engineering Technology and Sciences. World Journal of Advanced Engineering Technology and Sciences, 2023, 09(02), 351–358 Article DOI: https://doi.org/10.30574/wjaets.2023.9.2.0233*.

9. **Awua Justin Tarka** and Umar Abaka. (2023). Energy and Exergy Analysis of a Mixed Mode Solar Dryer Without Using Storage Material.. *International Journal for Science and Research Archive eISSN:2582-8185 Cross Ref DOI: 10.30574*/ijsra. Journal Home page: https//ijsra.net/

10. **J. T. Awua,** J. S. Ibrahim, Suseel Jai Krishnan, A. O. Edeoja, A. Kuhe, M. Sharifpur, S. M. S. Murshed. (2024). Synthesis, characterization, physicochemical, and electricalproperties of natural (bio) nanofluids. *Environmental Progress and Sustainable Energy*. *https://doi.org/10.1002/ep.14397*.

11. **J. T. Awua.,** J. S. Ibrahim., Suseel Jai Krishnan., A. O. Edeoja, A. Kuhe, M. Sharifpur, S. M. S. Murshed. Synthesis, characterization, physicochemical, and electrical properties of natural (bio) Nanofluids. *Environmental Progress and Sustainable Energy.* **https://doi.org/10.1002/ep.14397**

12. **Awua Justin Tarka** , Enyejo Lawrence Anebi and Mohammed Kabiru Bobby (2023) Evaluation of physical and combustion characteristics of briquettes of palm fruit fibre and maize chaff using cassava extract as binder.*World Journal of Advanced Engineering Technology and Sciences* DOI: https://doi.org/10.30574/wjaets.2023.9.2.0233

13. **Awua Justin.,** Kuhe Aondoyila., Archigbenda Victor, Agada Alexander., Adama Lukman. Assessment of the Greenhouse Gas Emissions Patterns of Hotels in Abuja, Nigeria. *International Journal of Engineering Technology Research & Management*. **https://www.ijetrm.com/** ISSN: 2456-9348.

14. **Awua Justin**, Kuhe Aondoyila, Okan Williams, Agada Alexanda, Shaka Onome.(2025) International Journal of Advances in Scientific Research and Engineering (ijasre) Volume-09 Issue 05**. DOI:** 10.31695/IJASRE.2025.6.4**. https://ijasre.net/**

15. **Awua JustinTarka ,** Jacob Sunday Ibrahim, Joseph Egbe , Archigbenda Victor**.** Performance and Emission Characteristics of Compression Ignition Engines Fuelled with Rubber Seed Oil Biodiesel Blended with Ethanol and Diethyl Ether. *Journal of Energy Research and Reviews*. Manuscript number: 2025/JENRR/138324.

Conference Proceedings

1. **Awua, J.T., Ibrahim** J.S., Edeoja, A. (2016). Investigation of the Thermal Conductivity of Nanofluid prepared from Palm Kernel Fibre Nanoparticles using De-ionized Water and Ethylene Glycol (60:40) as Base Fluid. *Nano Research Group (NRG), University of Nsukka, Enugu State, Nigeria.*
2. **J.T.Awua**, J.S. Ibrahim, A. Kwaghger, M. Sharipur and J.P. Meyer, (2016). Investigation into Thermal Conductivity of Palm Kernel Fibre Nanofluid with Mixture of Ethylene Glycol and Water as Base Fluid.. *12th International Conference on Heat Transfer, Fluid Mechanics and Thermodynamics (HEFAT 2016) in Malaga, Spain from 11th to 13th.*

SPECIAL SKILLS

Good in microsoft office application

Works moderately with ANSYS fluent CFD software

EXTRA CURRICULAR ACTIVITIES

Watching football and strolling

REFEREES

Engr. Prof. Jacob Sunday Ibrahim
Department of Mechanical Engineering
Federal University of Agriculture, Makurdi
Benue State
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